

NANO TECHNOLOGY

Advanced Professional Program · 22 Sessions · 6 Capstone Projects

PROFESSIONAL TRACK

VIRTUAL INTERNSHIP

WIPRO DICE ID ACCREDITED



Nanoscale Metrology



Quantum Devices



Nano Biomaterials



Nanorobotics & AI

OUR ALUMNI WORK AT

Microsoft

Capgemini

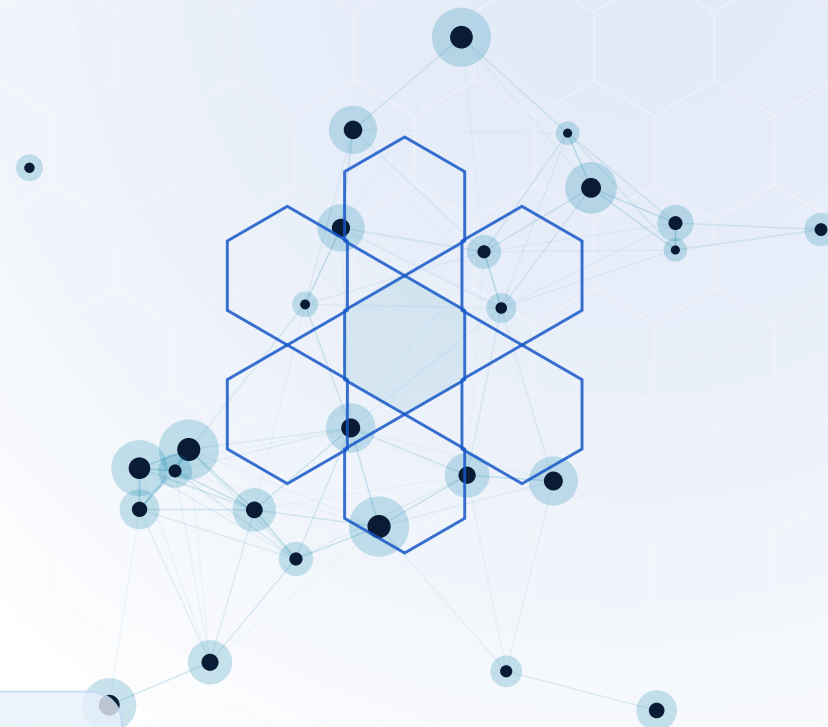
Deloitte

IBM

Adobe

EY

Google Cloud



Engineering Careers at the Nanoscale

At Edufyi Tech Solutions we transform the educational journey of college students through cutting-edge virtual internships and deep-tech training. Every learner works under industry mentorship and ships six real-time and capstone projects per domain — built on live lab data, simulation toolchains and publication-grade documentation.

Our Nano Technology track has been rebuilt for 2026: it now spans classical nanoscience fundamentals through quantum materials, nanophotonics, nano-bio interfaces, AI-accelerated materials discovery and semiconductor device fabrication — the exact stack hiring teams at research labs, fabs and deep-tech startups screen for.

**22**

Structured sessions across 6 learning tracks

**06**

Capstone & real-time industry projects

**150+**

Hiring partners for internship placement

**24×7**

Mentor support and lifetime LMS access

THE SCALE OF THE CRAFT



18 Reasons Learners Pick Edufyi



Self-Paced Mastery

Choose your speed, master the skills



Mentors from MNCs

Scientists, engineers & startup founders



Unlimited Mocks & GDs

Technical, HR and group discussions



150+ Internship Partners

Placement into real engineering teams



Bootcamps & Webinars

Certified deep-dive intensives



Performance-Based Stipend

Skill up and cash in



Industry Project Playground

Live problem statements from R&D teams



100% Placement Assistance

Referrals, mocks and interview loops



24x7 Dedicated Support

Never blocked on a doubt again



1:1 Counselling Sessions

Personalised career roadmapping



Lifetime Personalised LMS

Progress analytics that stay with you



Live Expert Mentorship

Weekly interactive masterclasses



Resume & Aptitude Grooming

ATS-ready profiles + aptitude drills



Exclusive R&D Content

Leading MNC research learning modules



Wipro DICE ID Certificates

Globally recognised accreditation



MNC Add-on Certifications

Microsoft, Cisco, IBM, Adobe & more



Recorded Lectures Anytime

Revisit every session on demand



Hiring Partner Access

Direct pipelines to recruiters

Track A — Foundations

S1

Induction & Lab Onboarding

Program roadmap, LMS walkthrough, simulation toolchain setup (COMSOL / LAMMPS / Quantum ESPRESSO trial stacks), safety and scientific-writing standards.

S2

Introduction to Nanoscience & Nanotechnology

Fundamentals of nanotechnology, history of nanoparticles, why size matters, surface-to-volume ratio, quantum confinement and the top-down vs bottom-up paradigms.

S3

Solid State Physics for Nanoscale Engineers

Crystallography, reciprocal lattice, scattering experiments, electron waves in crystals, band theory, superconductivity and magnetism.

S4

Advanced Tools for Nanoscience

Cleanroom practice, vacuum science, cryogenics and the instrumentation ecosystem behind modern nanoscale research.

S5

Nanoscale Measurement Techniques

Light and electron microscopy (SEM/TEM), scanning probe microscopy (AFM/STM), optical spectroscopy and diffraction-based characterisation.

Tracks are cumulative — each session ends with a graded lab notebook entry reviewed by your mentor.

Track B — Materials & Bio

S6

Nano Materials

How size, surface area and nanoscale forces give rise to a rich variety of materials: nanoparticles, nanowires, thin films, aerogels and metamaterials.

S7

Nanomaterials Synthesis

Sol-gel, CVD, ALD, hydrothermal and self-assembly routes; templating, precursors, mesoporous materials and their industrial applications.

S8

Nano Biomaterials — Part 1

Self-assembly principles in biology, biomimetic design, implant surfaces and tissue-engineering scaffolds at the nanoscale.

S9

Nano Biomaterials — Part 2

Targeted drug and gene delivery, encapsulation chemistry, theranostics and real-life clinical translation of nanomedicine.

S10

Nanotoxicology & Responsible Nano

How nanoparticles enter the body, cellular and tissue barriers, dose metrics, risk assessment, regulation and sustainable life-cycle design.

Track B includes a wet-lab simulation project on nanoparticle drug-carrier design.

Track C — Devices & Quantum

S11

Micro / Nano Electromechanical Systems (MEMS & NEMS)

Sensors, actuators, resonators, energy harvesters and the mechanics of moving structures at micro and nano scales.

S12

Nanofabrication Techniques

Photolithography, EUV, e-beam and nanoimprint lithography, etching, deposition and yield engineering in a modern fab.

S13

Semiconductors & Advanced Nodes

Doping, junctions, MOSFET scaling, FinFET and gate-all-around nanosheet architectures, and the physics behind Moore's-law successors.

S14

Device Physics

Engineering aspects of semiconductor nanomaterials, transport phenomena, contacts, reliability and device fabrication workflows.

S15

Nano-Optics: Theory & Practice

Near-field optics, photonic crystals, plasmonics, metasurfaces and nanoscale light-matter interaction.

Track C is delivered with fab-floor case studies from semiconductor hiring partners.

Track D — Frontier & Career

S16

Quantum Science

Cryogenics, superposition, entanglement, interference, qubit modalities and the quantum-materials pipeline.

S17

2D Materials & Heterostructures

Graphene, TMDCs, MXenes, hBN, twistronics and van der Waals stacking for next-generation electronics.

S18

Nanoelectronics for AI Hardware

Memristors, spintronics, neuromorphic crossbars and in-memory computing for energy-efficient AI accelerators.

S19

AI-Driven Materials Discovery

Machine-learning interatomic potentials, DFT-plus-ML screening, generative materials models and lab automation.

S20

Nano for Energy & Environment

Perovskite and tandem solar cells, solid-state batteries, catalysis, hydrogen storage, water purification membranes.

S21

DNA Nanotech & Molecular Machines

DNA origami, molecular motors, nanorobotics and programmable matter at the biology-engineering interface.

S22

Career Launchpad & Support

Interview tricks and tips, most-preferred technical questions, resume and cover-letter preparation, portfolio review.

Frontier Topics You Won't Find in a Basic Syllabus



Twistronics & Moiré Physics

Magic-angle graphene, flat bands, correlated superconductivity.



Plasmonics & Metasurfaces

Flat optics, SERS sensing, sub-wavelength beam shaping.



Quantum Dots & Single-Photon Sources

Colloidal QDs, QLED displays, quantum communication emitters.



Nanorobotics & Micro-Swimmers

Magnetically steered carriers for targeted therapy.



Spintronics & Magnonics

MRAM, spin-orbit torque, low-power memory logic.



Digital Twins & Multiscale Simulation

DFT → MD → FEM coupled modelling workflows.



Self-Healing & Smart Coatings

Stimuli-responsive, antifouling and superhydrophobic surfaces.



Lab-on-Chip & Nanofluidics

Microfluidic diagnostics, point-of-care biosensing.



Nano Standards, IP & Safety

ISO/TC 229, patents, EHS and responsible innovation.



Cryo-EM & Atom Probe Tomography

Atomic-resolution structural and compositional mapping.



Additive Nano-Manufacturing

Two-photon polymerisation and roll-to-roll nanoprinting.



Space & Defence Nanomaterials

Radiation-hard coatings, aerogels, nano-satellite structures.

The Stack You Will Actually Operate



Characterisation

- SEM / FE-SEM
- TEM & HRTEM
- AFM / STM
- XRD & XPS
- Raman & FTIR
- Cryo-EM basics



Fabrication

- Spin coating
- CVD & ALD
- E-beam lithography
- Sputtering
- RIE etching
- Cleanroom protocol



Simulation

- COMSOL Multiphysics
- Quantum ESPRESSO / VASP
- LAMMPS MD
- Lumerical FDTD
- MATLAB / Simulink
- ANSYS



Data & AI

- Python & NumPy
- scikit-learn
- PyTorch
- Materials Project API
- ImageJ / Gwyddion
- Origin & LaTeX

Every tool is taught through guided labs, recorded walkthroughs and a graded deliverable — not slides.

Six Capstones. Portfolio Proof.



Green Synthesis of Silver Nanoparticles

01

Synthesis, UV-Vis and XRD characterisation, antibacterial assay and a full lab report.



Perovskite Solar Cell Efficiency Study

02

Device stack simulation, defect analysis and optimisation of a tandem architecture.



Targeted Nano Drug Delivery Carrier

03

Liposome / polymeric carrier design with release-kinetics modelling and toxicity screening.



Graphene Field-Effect Biosensor

04

2D-material channel design, sensitivity modelling and a working detection prototype pipeline.



ML Model for Nanomaterial Property Prediction

05

Dataset curation from Materials Project, feature engineering and a deployed prediction notebook.



Plasmonic Metasurface Lens Design

06

FDTD simulation, nanoantenna array optimisation and a fabrication-ready layout.

Each capstone ships with a graded quiz, mentor code review, GitHub-ready documentation and a demo video.

WHERE THIS TAKES YOU

Careers, Industries & Growth

ROLES YOU CAN TARGET

- Nanotechnology Research Associate
- Semiconductor Process Engineer
- Materials Characterisation Analyst
- MEMS / Sensor Design Engineer
- Nano-Biotech R&D Associate
- Thin Film & Coatings Engineer
- Computational Materials Scientist
- Quantum Devices Trainee Engineer
- Failure Analysis Engineer



6

Portfolio projects on your resume



4+

Stackable industry certifications

INDUSTRIES HIRING



Semiconductors & Fabs



Pharma & Medical Devices



Energy & Batteries



Aerospace & Defence



Research Labs & Academia



Advanced Coatings & Materials



150+

Internship & hiring partners

Certificates That Recruiters Recognise

EDUFYI TECH SOLUTIONS

INTERNSHIP COMPLETION

presented to

SAMPLE CERTIFICATE

in recognition of outstanding completion of the program

 WIPRO DICE ID: 896354

Powered by Wipro DICE ID · blockchain-verifiable

EDUFYI TECH SOLUTIONS

PROJECT COMPLETION

presented to

SAMPLE CERTIFICATE

in recognition of outstanding completion of the program

 WIPRO DICE ID: 896354

One per capstone, with project title and duration

EDUFYI TECH SOLUTIONS

INDUSTRIAL TRAINING

presented to

SAMPLE CERTIFICATE

in recognition of outstanding completion of the program

 WIPRO DICE ID: 896354

Domain internship credential for academic credit

PROFESSIONAL MNC CERTIFICATIONS ALSO INCLUDED

Microsoft Azure Fundamentals

Cisco Cybersecurity

IBM Internet of Things

Autodesk Certified User

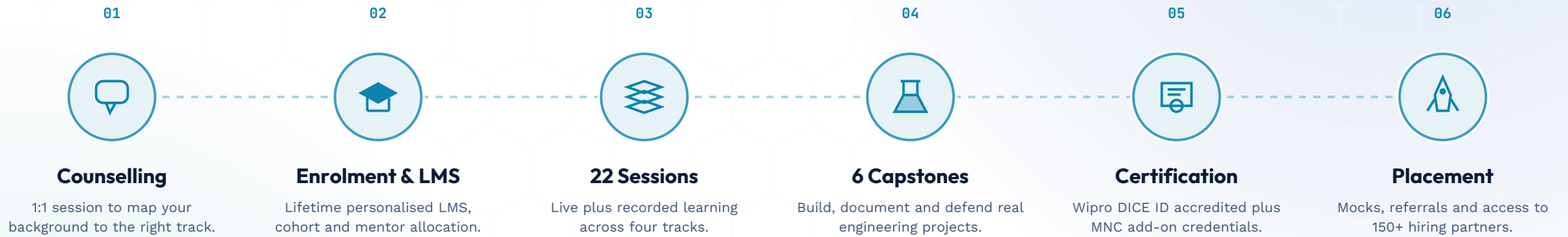
Adobe Certified

Intuit ESB

QuickBooks

All certificates are globally recognised, shareable on LinkedIn and verifiable by employers.

From Enrolment to Offer Letter



Eligibility & Format

Open to B.E / B.Tech / M.Tech / B.Sc / M.Sc students in Physics, Chemistry, Materials, ECE, EEE, Mechanical and Biotech.
100% virtual · flexible weekday and weekend batches · no prior nanotechnology experience required.

ACCREDITED CERTIFICATES

Certificates You Earn

Every learner receives government and industry aligned certificates, verifiable by employers and shareable on LinkedIn.



Internship Completion

Awarded on successful completion of the industrial internship.



Project Completion

Issued for every capstone project built and defended.



Industrial Training

MSME · ISO · NSDC aligned industrial training credential.

VERIFIED CREDENTIALS

Wipro DICE ID accredited · MSME & ISO aligned · QR verifiable · Recognised by 150+ hiring partners

Global MNC Certifications

Learners are prepared for and certified through globally recognised professional certification programs from leading technology companies.



CISCO

Cybersecurity CCST



MICROSOFT

Azure Fundamentals



IBM

Internet of Things



AUTODESK

Certified User



ESB

Entrepreneurship

Authorised Partners

Edufyi Tech Solutions is an authorised certification and testing partner for the world's leading technology certification bodies.



ADOBE



MICROSOFT



QUICKBOOKS



AUTODESK



CISCO



ESB

AUTHORISED TESTING & CERTIFICATION CENTRE

Certify on campus with globally accepted credentials.

Certify. Build. Get Hired.



Accredited Certification



Microsoft Azure Fundamentals



Cisco Cybersecurity CCST



IBM Internet of Things



Adobe Certified Professional



Autodesk Certified User



ESB Entrepreneurship



QuickBooks Certified User

START YOUR JOURNEY

Enrol in the specialised Industrial Training & Internship program with 150+ partnered companies. Learn from industry experts, ship 6+ real-time projects and earn globally recognised certifications.

WEB www.edufytechsolutions.com

MAIL edufyi@edufytechsolutions.in

CALL **+91 63620 00213**

EDUFYI TECH SOLUTIONS